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Eco-design of Magnetic Components in Power Electronics: A Life Cycle Assessment Perspective

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Abstract

This paper presents a comparative life cycle assessment (LCA) of inductors for photovoltaic (PV) inverters, with particular focus on two designs optimized for conflicting objectives, namely minimum weight and minimum power loss. The carbon emissions associated with the manufacturing stage are primarily determined by the component weight, whereas those in the use stage are governed by the power loss during operation. As a result, a fundamental trade-off arises, and its outcome depends strongly on the operating lifetime of the component. The results show that the minimum-weight design is more favorable for short-lifetime applications, while the minimum-loss design leads to a lower overall carbon footprint when the operating lifetime exceeds a break-even point. These findings suggest that operating lifetime should be explicitly considered as a key criterion in the sustainable design of magnetic components.

1 Introduction

The global commitment to carbon neutrality is shifting the priorities of power electronics from purely efficiency- and power-density-oriented design toward a broader life-cycle perspective [1], [2]. In parallel, sustainability assessment is increasingly being discussed not only at the device and converter levels, but also at the PV-system level, where the environmental burden of constituent components can no longer be neglected [3], [4], [5]. As the research background of this study, Fig. 1 illustrates a typical three-phase grid-connected inverter with an L-type filter, which serves as a representative platform for analyzing the environmental implications of magnetic design.

In such grid-connected systems, the interface stage must simultaneously satisfy current-quality requirements, support stable grid interaction, and maintain satisfactory dynamic performance under varying operating conditions [6], [7], [8]. Consequently, passive filter components, especially output inductors, remain indispensable in practical converter implementation. Their electrical function is well recognized, but their material demand, manufacturing burden, and use-phase loss contribution also make them non-negligible sources of embodied and operational carbon emissions. This issue becomes particularly relevant in PV inverters, where long service lifetimes amplify the cumulative impact of component-level design decisions.

From a design perspective, conventional magnetic and filter synthesis is generally formulated as a multi-objective optimization problem, in which power loss, mass, volume, cost, and thermal constraints must be balanced [9], [10], [11]. For PV and grid-connected inverters, this research line includes classical filter design methods [12], optimum inductance-capacitance allocation [13], and size- and thermal-aware high-frequency implementations [14]. More recently, machine-learning-assisted optimization has also been introduced into inverter design workflows to improve design efficiency and expand the searchable solution space [15]. These studies have significantly improved electrical and physical design quality, yet their optimization objectives are still dominated by technical performance indicators.

At the same time, broader sustainability-oriented

studies in power electronics have started to quantify environmental impacts at the converter or system level, including inverter-level life cycle assessment, semiconductor technology comparisons, and roadmap-style assessment frameworks [5], [16], [17]. These works clearly show that environmental impact cannot be inferred from efficiency alone, and that embodied emissions from material extraction, processing, and manufacturing may be comparable to or even dominate use-phase impacts under certain scenarios. However, such insights have only rarely been translated into component-level magnetic design rules for PV inverters, where geometric and material choices directly determine both manufacturing footprint and operating losses.

In parallel, the wider power electronics community is rapidly adopting advanced design, control, and reliability methodologies. Representative examples include ANN-aided parameter design and PSO-ANN-assisted control strategies [18], [19], comprehensive reviews of efficient and reliable converter control [20], and reliability-oriented condition-monitoring techniques for power devices [21]. Although these methods demonstrate the growing role of data-driven and reliability-aware thinking in modern converter engineering, they are still seldom linked to life-cycle environmental metrics when selecting magnetic materials, geometries, and operating points.

A key unresolved issue is therefore the tradeoff between manufacturing-related and use-phase carbon emissions. A design optimized purely for minimum loss often requires more material and larger magnetic structures, thereby increasing embodied carbon. In contrast, a lightweight design may reduce manufacturing emissions but lead to higher long-term operational losses. This tradeoff is further complicated by the fact that electricity-related carbon intensity is not constant over time. Foundational and recent studies on electricity modeling and dynamic life cycle assessment have shown that time-varying grid emission factors can substantially affect the environmental ranking of alternative designs [22], [23], [24]. As a result, a design that appears environmentally favorable under a static emission factor may not remain optimal across the full service life of the converter.

To address this gap, this paper introduces an eco-design framework that explicitly integrates Life Cycle Assessment (LCA) into magnetic component design. Focusing on the output filter of a 10 kW

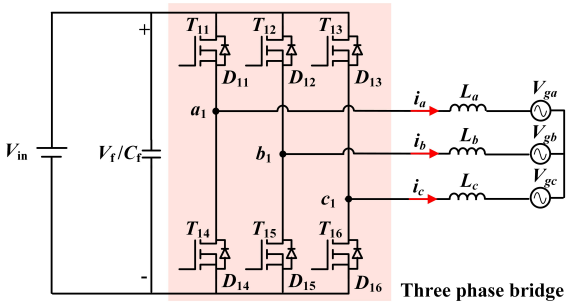


Fig. 1: Three phase inverter with L filter.

PV inverter, we compare two practical optimization targets: minimum weight and minimum power loss. By combining detailed electrical models with LCA data, this study evaluates how these conflicting objectives influence the total environmental impact, providing a quantitative basis for sustainable magnetic design..

The main contributions of this paper are summarized as follows:

- A comprehensive eco-design framework is proposed, integrating electrical sizing constraints with cradle-to-grave LCA data for magnetic components.
- A comparative life-cycle carbon footprint analysis is conducted between weight-optimized and loss-optimized filter inductors, revealing a critical, lifetime-dependent environmental break-even point.

2 Design Methodology and Life Cycle Assessment Framework

This section details the iterative sizing algorithm for the magnetic components, the simulation verification, and the proposed LCA framework.

2.1 Inductor Sizing Algorithm and Optimization Objectives

The design of the filter inductor is primarily governed by the requirement to maintain the grid-connected total harmonic distortion (THD) below 3% to ensure high power quality. For the considered three-phase grid-tied PV inverter, the system operates with an 800 V DC-link voltage (V_{dc}), a 230 V AC grid phase voltage, and a 10 kHz switching frequency (f_{sw}).

The relationship between the maximum peak-to-peak current ripple $\Delta i_{L,\max}$, the inductance L, and the THD limit can be expressed as the following

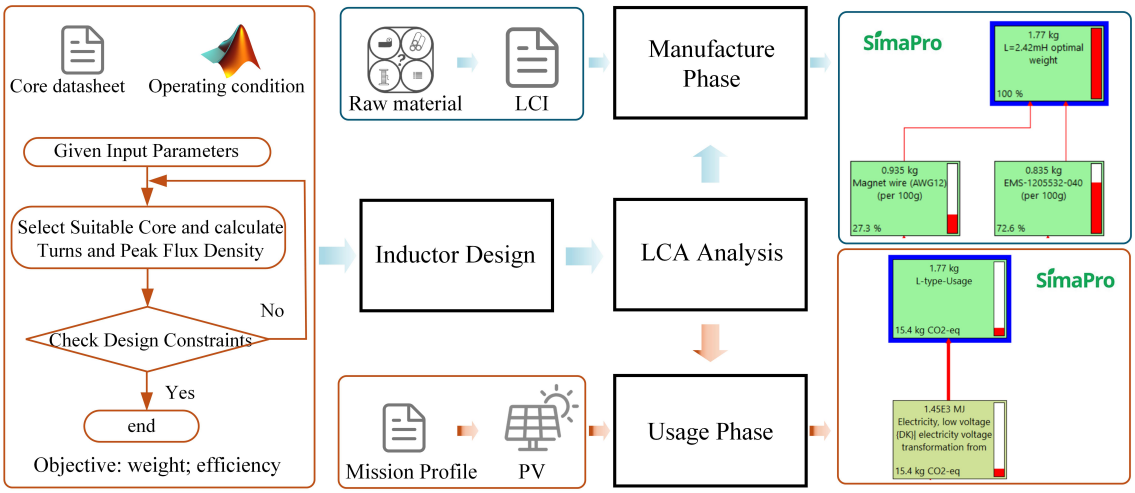


Fig. 2: Integrated magnetic design and LCA framework.

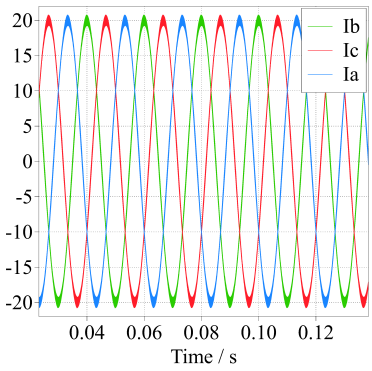


Fig. 3: Simulated Grid Current and THD of 2.54%.

constraint group:

$$\begin{cases} \Delta i_{L,\max} = \dfrac{V_{\mathrm{dc}}}{8 f_{\mathrm{sw}} L} \\ \mathrm{THD}_i < 3\% \end{cases} \tag{1}$$

Based on (1), the target inductance is selected as $L = 2.42$ mH. To verify the effectiveness of this design, a PLECS simulation was conducted under full-load conditions. As shown in Fig. 3, the simulated grid current waveform exhibits a THD of 2.54%, which successfully satisfies the sub-3% grid-code requirement.

The physical realization of the 2.42 mH inductor follows a well-established, iterative procedure utilizing a manufacturer core database (e.g., Micrometals), as illustrated in Fig. 4.

The algorithm selects candidate cores from the database and strictly verifies that they satisfy the following boundaries:

1) Core Saturation Constraint The peak magnetic flux density $B_{\max}$ must not exceed the saturation limit B_{sat} of the selected material to avoid a drop in inductance:

$$B_{\max} = \frac{L I_{\mathrm{peak}}}{N A_{\mathrm{e}}} \leq B_{\mathrm{sat}} \tag{2}$$

where I_{peak} is the peak inductor current, N is the number of turns, and A_{e} is the effective cross-sectional area of the core.

2) Winding Window Fill Factor Constraint The selected core must have sufficient window area to accommodate the copper windings. The window fill factor K_{u} is restricted by:

$$K_{\mathrm{u}} = \frac{N A_{\mathrm{w,bare}}}{W_{\mathrm{A}}} \leq K_{\mathrm{u,max}} \tag{3}$$

where $A_{\mathrm{w,bare}}$ is the bare cross-sectional area of the wire, W_{A} is the available window area of the core, and $K_{\mathrm{u,max}}$ is the empirical maximum fill factor.

3) Thermal Constraint The total power loss (copper loss plus core loss) must be dissipated effectively to ensure the temperature rise ΔT stays within predefined thermal limits. For each candidate design, the thermal feasibility is evaluated from the total power loss,

$$P_{\mathrm{tot}} = P_{\mathrm{cu}} + P_{\mathrm{core}} \tag{4}$$

where P_{cu} and P_{core} denote the copper loss and core loss, respectively.

The copper loss is determined by the RMS current flowing through the winding and the corresponding AC resistance:

$$P_{\mathrm{cu}} = I_{L,\mathrm{rms}}^2 R_{\mathrm{ac}} \tag{5}$$

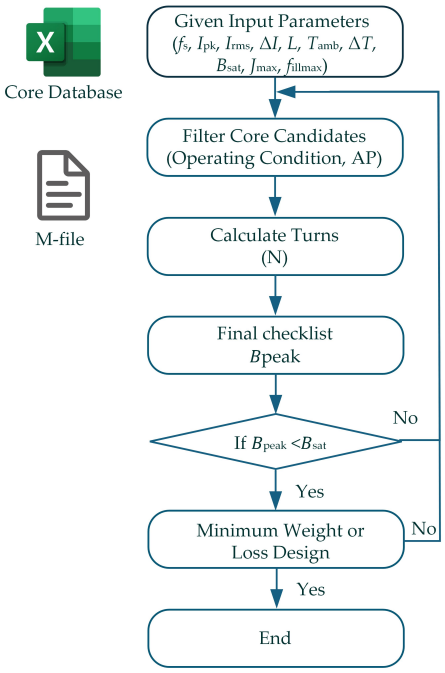


Fig. 4: Inductor sizing flowchart for weight/loss optimization

where $I_{L,\mathrm{rms}}$ is the RMS inductor current. The winding resistance can be expressed as

$$R_{\mathrm{ac}} = k_{\mathrm{ac}} R_{\mathrm{dc}}, \qquad R_{\mathrm{dc}} = \rho(T)\frac{l_{\mathrm{cu}}}{A_{\mathrm{cu}}} \tag{6}$$

where k_{ac} is the AC resistance factor accounting for skin and proximity effects, $\rho(T)$ is the temperature-dependent resistivity of copper, l_{cu} is the total conductor length, and A_{cu} is the effective copper cross-sectional area. For the initial design screening, R_{ac} can be approximated by R_{dc} if the AC effect is not dominant; otherwise, a frequency-dependent correction factor is introduced.

The core loss is estimated from the switching frequency and magnetic flux swing using a Steinmetz-type model:

$$P_{\mathrm{core}} = k f_{\mathrm{sw}}^{\alpha} B_{\mathrm{ac}}^{\beta} V_{\mathrm{core}} \tag{7}$$

where k, α, and β are material-dependent coefficients obtained from the core manufacturer data, B_{ac} is the AC flux density amplitude, and V_{core} is the effective core volume. When manufacturer loss-density data are directly available, P_{core} can also be evaluated from the tabulated loss density under the corresponding $(f_{\mathrm{sw}}, B_{\mathrm{ac}})$ operating point.

Based on the total loss, the temperature rise is approximated by a lumped thermal resistance model:

$$\Delta T = P_{\mathrm{tot}} R_{\mathrm{th}} \leq \Delta T_{\max} \tag{8}$$

where R_{th} is the equivalent thermal resistance from the inductor to ambient, and $\Delta T_{\max}$ is the allowable temperature rise. Therefore, only the candidate designs satisfying (4)–(8) are retained in the subsequent optimization process.

Applying this framework to the 2.42 mH requirement, the resulting design parameters for the two optimization branches are summarized in Table 1. The minimum-mass design successfully selects a more compact LE114 core, while the minimum-loss design opts for a larger LE130 core with an increased copper area to minimize winding resistance and thermal dissipation.

2.2 Life Cycle Assessment Framework

To quantify the environmental implications of the aforementioned design choices, a standardized LCA is incorporated into the design loop. The evaluation framework is structured into the following key phases:

1) Goal and Scope The primary objective is to compare the life-cycle carbon footprint of the weight-optimized inductor versus the efficiency-optimized inductor, ensuring both designs deliver identical functionality for a 10 kW PV inverter.

2) Functional Unit In LCA, the functional unit serves as the basis for all calculations. Here, it is defined as the provision of the specified 2.42 mH filter inductance for a 10 kW three-phase PV inverter operating under a defined mission profile.

3) System Boundaries This study adopts a streamlined cradle-to-grave approach focusing on the two dominant stages: the Manufacturing phase (accounting for embodied carbon in raw materials and production) and the Use phase (accounting for operational emissions resulting from power losses). Environmental impacts stemming from the installation, transportation, and end-of-life (EoL) phases are currently excluded.

3 Results and Discussion

This section instantiates the proposed LCA framework using SimaPro software (version 9.3) for the 10 kW PV case, applying the same functional unit, system boundaries, and impact assessment method to both inductor designs.

Tab. 1: Comparison of Optimal Inductor Designs

Parameter	Minimum Mass Design	Minimum Loss Design
Core model	LE114, 00K114LE026	LE130, 00K130LE026
Turns (N)	118	113
Winding	6 $\times$ 1.0 mm parallel	10 $\times$ 1.0 mm parallel
Copper area	4.71 mm^2	7.85 mm^2
Fill factor	0.428	0.453
Flux density	0.424 T	0.479 T
Loss	22.05 W	16.59 W
Weight	1.17 kg	2.81 kg

3.1 Life Cycle Inventory and Mission Profile

To accurately assess the manufacturing-phase embodied carbon, a detailed Life Cycle Inventory (LCI) was compiled for both inductor designs based on their physical realization. Fig. 5 illustrates the material composition by weight for the two candidate filters.

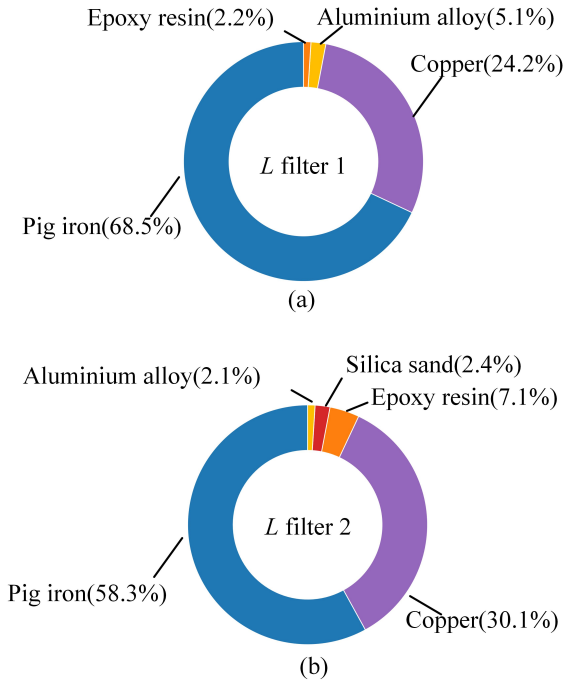


Fig. 5: Material composition breakdown for the Life Cycle Inventory (LCI). (a) Minimum Mass Design. (b) Minimum Loss Design.

As shown in Fig. 5(a), the minimum-mass design is predominantly composed of pig iron (68.5%) for the magnetic core and copper (24.2%) for the windings. It also includes minor shares of aluminium alloy (5.1%) for mounting structures and epoxy resin (2.2%) for insulation. Conversely, to achieve higher efficiency, the minimum-loss design utilizes a larger LE130 core and a significantly thicker winding structure. As depicted in Fig. 5(b), this shifts the mass distribution to 58.3% pig iron and 30.1% copper. Furthermore, it incorporates a higher proportion of potting materials, including epoxy resin (7.1%) and silica sand (2.4%), to enhance thermal dissipation. These specific material mass breakdowns were mapped to the Ecoinvent database (version 3.8) to calculate the total embodied emissions. The Global Warming Potential (GWP), expressed in kilograms of $\mathrm{CO_2eq}$, was calculated using the IPCC 2021 characterization factors. For the use-phase evaluation, a measured PV mission profile derived from a typical meteorological year (TMY) was applied to capture the dynamic solar irradiance and load conditions. The power loss of each inductor is integrated over this profile and multiplied by the respective grid carbon intensity factor to quantify the operational carbon footprint per year. The power loss of each inductor is integrated over this profile and multiplied by the respective grid carbon intensity factor to quantify the operational carbon footprint per year.

3.2 Life-Cycle Carbon Trade-off and Break-Even Analysis

The application of the proposed framework yields insightful results regarding the environmental trade-offs inherent in magnetic component design. Fig. 7 synthesizes the life-cycle carbon footprint analysis for the two design strategies.

Fig. 7(a) illustrates the manufacturing phase's embodied $\mathrm{CO_2eq}$ emissions versus inductance L. A positive correlation exists: increasing L necessitates more core and copper materials, raising embodied emissions. At any given inductance, the optimal-weight design consistently exhibits lower manufacturing impact. Conversely, the optimal-loss

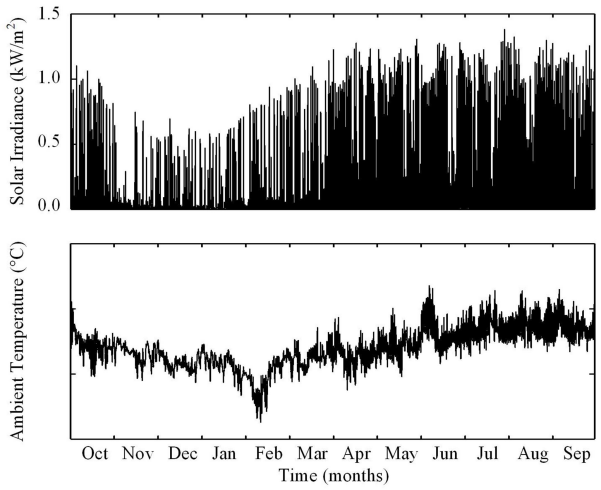


Fig. 6: Yearly PV mission profile used for the use-phase evaluation.

design requires larger cores and thicker conductors to minimize resistance, resulting in a higher initial "carbon investment" that can be 20–40% greater than its mass-optimized counterpart.

Conversely, the operational phase presents a reversed trend, as shown in Fig. 7(b). The loss-optimized design significantly reduces annual $\mathrm{CO_2eq}$ emissions by minimizing heat-related energy waste under the specified PV mission profile. Although emissions for both designs increase with inductance, prioritizing efficiency yields a lower carbon footprint over the operational lifespan.

The synthesis of these opposing trends for $L = 2.42$ mH is illustrated in Fig. 7(c), revealing a dynamic "environmental break-even point." Initially, the weight-optimized design is superior due to lower embodied emissions. However, as operational losses accumulate linearly, a crossover occurs at approximately 5.4 years. Beyond this threshold, the loss-optimized design becomes environmentally superior as operational savings eventually offset the higher initial manufacturing impact.

These findings indicate that sustainability is context-dependent rather than universal. While minimizing weight is more effective for short-term deployments, prioritizing efficiency provides a greater net environmental benefit for typical 20–25 year PV installations.

4 Conclusion

This paper presented an integrated life-cycle-oriented framework for evaluating the environmental impact of magnetic components in power electronics. By applying the framework to the output filter inductor of a 10 kW PV inverter, two practical design philosophies, namely minimum weight and minimum power loss, were compared under the same electrical requirement and yearly PV operating profile. The results revealed a clear trade-off: the minimum-weight design reduces embodied carbon in the manufacturing stage, whereas the minimum-loss design lowers the operational carbon footprint during the use stage.

A time-dependent break-even point of approximately 5.4 years was identified in the analyzed scenario. Below this duration, the lightweight design provides a lower total carbon footprint, while above it, the high-efficiency design becomes the environmentally preferable option. These findings indicate that service lifetime should be treated as an explicit design criterion in the eco-design of magnetic components, rather than relying on a single fixed optimization target.

Future work will extend the present framework by incorporating additional life-cycle stages and by applying the methodology to other passive components and converter topologies.

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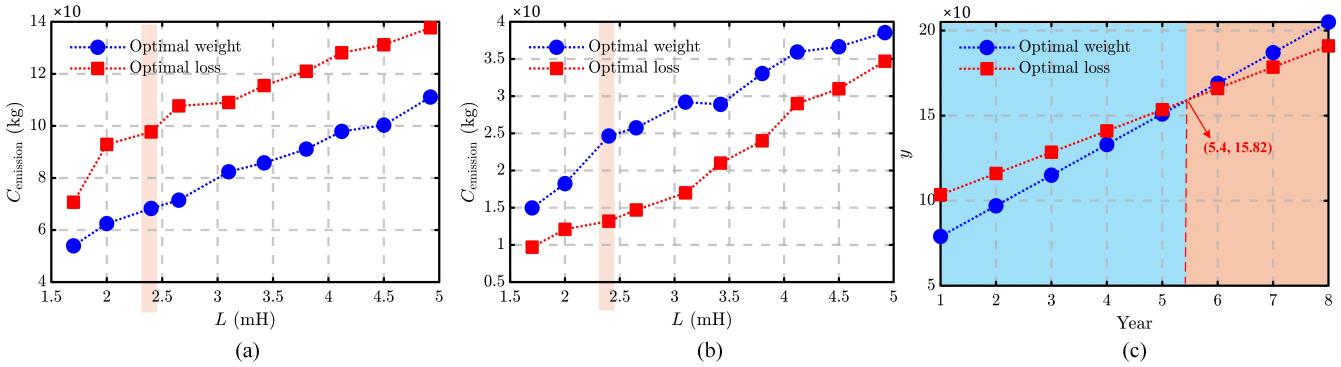


Fig. 7: Life-cycle carbon trade-off analysis. (a) Embodied $\mathrm{CO_2eq}$ from manufacturing versus inductance L. (b) Annual operational $\mathrm{CO_2eq}$ versus L. (c) Cumulative life cycle $\mathrm{CO_2eq}$ versus operation years for $L = 2.42$ mH, highlighting the break-even point.

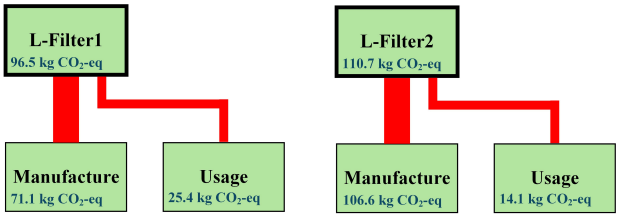


Fig. 8: SimaPro simulation results of carbon footprint for two inductor designs: (a) optimal mass design, and (b) optimal loss design.

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